HospitalOS

Hospital Operating System for Africa

HOSPITALOS DOCUMENTATION

Tenant Guide

Hospital staff reference

VERSION

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PREPARED BY

John Ipadeola, Production Team Lead

AgoroX Africa

1. Who this guide is for

This guide is for hospital staff using HospitalOS day to day — clinicians, diagnostic and pharmacy staff, finance and administration teams. It covers every module in the system and, where it matters, why a screen behaves the way it does.

Role	Scope	Account
Administrator	Your hospital's full HospitalOS deployment	Any account with the Super Admin role
Clinical staff	Patient care, diagnostics, pharmacy for your role	Doctor / Nurse / Lab Scientist / Radiographer / Pharmacist accounts
Finance staff	Billing, payments, claims	Cashier account

Two-level permissions. *Every role has an AREA list (which sidebar groups it can open) and a GRANTS list (which specific actions it can take once inside). Section 3 covers this in full, with the actual permission table.*

2. Signing in and how the system is organised

Sign in with your hospital email and password. An administrator sets your role under Administration → Users & roles when your account is created — there is no self-service sign-up.

Security. *Your password is checked server-side against a securely hashed copy — it is never stored or transmitted in readable form. Repeated failed sign-in attempts are automatically rate-limited.*

2.1 Signing back in on a shared device

A device remembers the last email signed in on it — the next time the sign-in screen loads there, that email is pre-filled and the cursor jumps straight to the password field, so signing back in only needs a password. This is a genuine convenience for shared hospital workstations (a nurses' station tablet, for instance) where several staff use the same device across a shift, and it changes nothing about the actual sign-in: the password is still required in full and still verified server-side, exactly as above.

Only the email is ever remembered — never a password, never anything that could grant access by itself. If a colleague's email is showing, “Not you?” next to it clears the remembered email on that device immediately.

2.2 Signing out

Signing out ends your session properly — it doesn't just clear it from this browser, it revokes the session on the server the moment you sign out. A token from a signed-out session is dead immediately and cannot be reused, even if it had time left before its automatic expiry. On a shared workstation, signing out genuinely closes your access rather than leaving a reusable session behind, which is why it's the right habit on any device more than one person uses.

2.3 The thirteen sidebar groups

The sidebar is arranged by workflow, roughly in the order work happens through a hospital, not by department hierarchy:

Group	What lives there
Overview	Dashboard, alerts, worklist, communication hub, online bookings, referrals
Patient care	Registration/ADT, wards, ICU, emergency, theatre, maternity, records, renal, and 21 specialist clinics
Diagnostics	Laboratory, blood bank, imaging, radiotherapy, biobanking, instruments gateway, diagnostic intelligence
Pharmacy	Dispensing, inventory, NAFDAC formulary
Specialty services	Nutrition, sickle cell centre, dental, IPC, social work, occupational health, chaplaincy
Finance & trade	Billing, payments, NHIA & insurance claims, bank reconciliation, procurement, stores
Operations	Scheduling, CSSD, biomedical equipment, facility & waste, fleet, support services, visitors
Academic	Training programmes, clinical logbooks, CME, research, ethics committee
Public health	Surveillance, immunisation (full NPHCDA schedule per child), outreach, national reporting
Intelligence	Analytics, reports, forecasting
Compliance	Practitioner licenses, accreditation, incident & risk management, policies & SOPs
Administration	Users & roles, facilities, instruments gateway, pricing, documents, security & audit, settings
Academy	Role-based professional certification — marked Locked, in development (see 17.2)

You only see what your role grants. *A pharmacist's sidebar shows Overview and Pharmacy; a cashier's shows Overview and Finance & trade. This is not secrecy — it keeps each person's view to what they actually use.*

3. Roles and permissions

Area permissions gate the sidebar and routes — enforced on the server for every request, not just hidden on screen. Action permissions gate the buttons within an area. Separation of duties is enforced, not just suggested:

Role	Can	Cannot
Nurse	Admit patients, record vitals, file clinical notes	Discharge a patient
Lab Scientist	Collect samples, enter results, verify results	Reach Patient care at all
Radiographer	Order and report imaging studies	Verify laboratory results
Pharmacist	Dispense medication, restock inventory	Reach Diagnostics at all
Cashier	Take payments, file insurance claims	Approve a claim

Role	Can	Cannot
Records Officer	Register new patients	Admit or discharge a patient
Doctor	Full patient-care and diagnostics actions	Administration functions
Super Admin	Everything	—

The last-admin protection. *The last active Super Admin account cannot be deactivated by anyone, including another Super Admin — this prevents a hospital from accidentally locking itself out of its own system. When any other account is deactivated, its live sessions are revoked at once, so a removed staff member is signed out immediately rather than able to keep working from an already-open tab.*

4. Overview and the alert system

The Dashboard opens on live occupancy, admissions, and revenue charts. Every stat card links directly into the module behind it.

4.1 Alerts — many sources, one feed

Source	Raises an alert when…	Severity
Laboratory	A result crosses a panic threshold	Critical
Critical care	Any ICU/HDU vital goes critical	Critical
Radiology	A report is flagged with an urgent finding	Critical
Maternity	A newborn's Apgar score is below 7	Critical
Blood bank	A group is below reorder / a unit nears expiry	Critical / Warning
Pharmacy	A drug is out of stock / below reorder	Critical / Warning
Instruments	A device errors or goes offline	Critical / Warning
Renal	A scheduled dialysis session is overdue	Critical
Operations	Equipment under repair / a vehicle is out of service	Warning
Oncology	A chemotherapy cycle is overdue	Warning
Public health	A notifiable disease is trending upward	Warning
Referrals	An Emergency-urgency inbound referral is logged	Critical
Privacy	A data-subject request is overdue against its statutory window	Warning

Most alerts clear themselves — they read live state, so fixing the underlying problem clears the alert automatically, nothing to dismiss.

4.2 Notifications

Current limitation. *The bell icon surfaces booking requests inside the app only. It does not yet push a notification outside HospitalOS — that requires the Communication hub's channels to be connected to a real SMS/email carrier, which isn't wired up yet. Its count does not shrink on its own; a booking only leaves the bell once someone actually confirms or declines it.*

5. Patient care

5.1 Registration, admission and the ward board

The bed board and ADT read the same registry, so a bed taken in one is unavailable in the other instantly — double-booking is not possible.

5.2 Accommodation tiers

Tier	Rate / night
General Ward	₦15,000
Semi-Private	₦35,000
Private Room	₦60,000
Private Suite	₦120,000
VIP Suite	₦220,000
Executive Suite	₦350,000
Critical Care	₦180,000

Catalogue defaults — every rate can be overridden for your hospital. See Section 15.

5.3 The medical record

SOAP-structured clinical notes, an ICD-10 problem list, and recorded allergies. Filed notes cannot be edited or deleted — a correction is filed as an amendment referencing the original, with both remaining visible. This is enforced on the server, not just in the interface.

Allergy safety is enforced, not advisory. *A severe allergy on a patient's record disables the Pharmacy dispense button outright for that drug — and if the allergy check itself fails to load, dispensing is blocked until it succeeds, rather than silently allowing a dispense with an unverified check.*

5.4 Referrals and bookings

Referrals in and out of your facility are tracked from first contact through acceptance or decline, through to the patient's arrival. Online bookings holds appointment requests. Confirming and checking in a booking, or accepting and checking in a referral, calls the same function Outpatient uses internally — a checked-in patient is a genuine addition to today's queue, not a separate list.

5.5 Specialist units

Unit	What it does
Renal & dialysis	Haemodialysis programme (vascular access, schedule, fluid removal) and a CKD registry that auto-stages from eGFR
ICU / HDU	Five vitals per bed with panic thresholds; a critical reading marks the patient Unstable and raises an alert automatically
Emergency	Sorts by triage acuity first, arrival time second
Maternity	Tracks labour through delivery; a newborn Apgar below 7 raises a critical alert immediately

Unit	What it does
Oncology	Chemotherapy cycle tracking; an overdue cycle raises an alert, not a quiet reschedule
Geriatric	Falls-risk and polypharmacy screening on admission
Mental health	Inpatient care with admission status, observation level, and risk-flag tracking
21 specialist clinics	Cardiology through Neurosurgery, filtered views inside one Specialist clinics module

6. Diagnostics

Laboratory tests across multiple disciplines, and imaging protocols across Ultrasound, CT, MRI, General Radiography, and Mammography.

6.1 Laboratory and blood bank

Orders move Ordered → Collected → Resulted → Verified; each result flags Low/High/Critical live against its reference range. Blood bank crossmatching is engineered so two simultaneous requests for a scarce blood type can never both be matched to the same physical unit — proven directly under concurrent load before shipping, not just assumed safe.

Blood units can be added in a batch: choose the group and expiry window, set a quantity, and each unit is recorded with its own sequential tag — a delivery of several identical units is one action, not many.

6.2 Instruments & devices gateway

Category	Protocol	Examples
Laboratory analyzer	HL7 v2 / MLLP	Sysmex, Cobas, and older analyzers still in service
Imaging modality	DICOM 3.0	CT, MRI, ultrasound, older CR readers
Radiotherapy system	DICOM-RT / proprietary	A modern LINAC and a manual Cobalt-60 unit
Printer	IPP / raw socket / serial	Label, report, and legacy dot-matrix printers

Current limitation. *Device management, status, and message history are real and persist. The live network listener itself — an MLLP socket, a DICOM SCP, a print daemon — is server/network territory a browser-based feature can't stand up on its own; every action in this screen calls the exact function a live listener would call.*

6.3 Biobanking & lab utilities

Biobanking tracks long-term specimen storage across four capacity-limited units, distinct from the active Laboratory worklist. Lab utilities holds clinical calculators, unit converters, and bench reference material.

7. Pharmacy

7.1 Dispensing

Before any dispense completes, two independent checks run: the patient's recorded allergies against the drug, and the requested quantity against genuine current stock. A severe allergy match disables the dispense action outright — there is no override click-through, and if the allergy check itself fails to load, dispensing is blocked entirely rather than silently allowed through an unverified check.

7.2 Stock, batches, and expiry

A dispense cannot exceed what's actually on hand, checked on the server at the moment of the request — not a possibly-stale count the screen happened to be showing. Restocking above the reorder threshold clears a low-stock alert automatically; there's nothing to manually dismiss once the real condition is fixed.

Each restock records a batch with an optional batch number and expiry date. A drug's total stock is the sum of its batches, so stock figures and low-stock alerts behave exactly as before. When a drug is dispensed, stock is drawn first-expiry-first-out — the soonest-expiring batch is used first, untracked stock last — so the shelf clears in the order that minimises waste. Batches that are expired or expiring soon surface as expiry alerts, so stock can be pulled before it is ever dispensed.

7.3 Prescriptions and interactions

Prescriptions are an optional queue between prescribing and dispensing: a clinician raises a prescription (patient, drug, quantity, dosage), and the pharmacy fulfils it from a pending queue. Fulfilment runs the identical dispense path — allergy check, stock guard, first-expiry-first-out drawdown, and patient charge all apply — so a prescription that would oversell is refused, not forced through. Direct dispensing still works unchanged; the queue is an addition.

Interaction warnings are decision support, not a block. *When a patient and drug are chosen, the drug is checked against what the patient is currently on (recent dispenses plus pending prescriptions) using a curated list of major interactions. A match is shown with the interacting drug, severity, and clinical concern — exactly like allergies are surfaced. A clinician who judges the combination appropriate can still prescribe.*

7.4 The formulary

Aligned to NAFDAC-registered products, so drug names, forms, and strengths match what your hospital is actually licensed to dispense in Nigeria.

8. Finance & trade

8.1 Billing

Charges aggregate from five sources automatically: laboratory, pharmacy, radiology, theatre, and accommodation. A theatre case becomes billable the moment it enters theatre, not when scheduled. A payment can never exceed what's genuinely owed — checked on the server against the hospital's own current prices, not just on screen.

8.2 Payment methods and cash sessions

Method	Settles
Cash	Immediately — the money is physically in hand
Card, POS, Bank Transfer, Online Payment	Starts pending; confirmed once the funds actually arrive
NHIA / HMO	Recorded when a claim is marked paid (see 8.4)

Taking a Cash payment requires an open cash session — open one at the start of a shift with a counted opening balance, close it at the end with a counted actual. Expected cash is calculated automatically from that session's own Cash payments and Cash refunds, and any variance is recorded, never hidden.

Online Payment is live. *Card and online checkout run through a real payment provider (Paystack) on this deployment. When a patient pays online, the confirmation isn't taken on trust from the browser — the provider's notification is signature-verified and the transaction is independently re-confirmed against the provider before the payment is ever recorded, so only genuinely completed payments post to the patient's account.*

8.3 Invoices, refunds, and chargebacks

An invoice is a permanent snapshot of charges, generated deliberately and separate from the live-recalculated running balance — once created, it cannot be edited. A refund is the hospital voluntarily giving money back; it creates a new record rather than touching the original payment, and a payment can never be refunded for more than what's left on it, even when two refunds are attempted at the same moment. A chargeback is different: it's raised by the card issuer, not the hospital, and only affects revenue once it's actually resolved — raising one just puts the dispute on record.

8.4 Claims and co-payments

Claims move Submitted → Approved/Rejected → Paid — a cashier can file a claim but cannot approve it, and the workflow itself blocks marking a claim paid before it has been approved. A claim can carry an optional co-payment, the portion the patient pays directly; collecting it is a separate action from filing the claim, since a co-pay is normally taken at the point of service. A co-payment can only be collected once — the system won't take it twice, even if the collect action is triggered more than once.

8.5 Bank reconciliation

A Bank Transfer payment is never treated as confirmed just because it was recorded — only a matching real bank transaction does that. Import a CSV statement from your bank; each transaction is checked against pending Bank Transfer payments, and anything ambiguous goes to a manual review queue rather than being guessed at.

9. Operations

The machinery that keeps a hospital physically running, seven distinct areas under one group.

Area	What it tracks
Scheduling & rosters	Staff shift rosters across departments
CSSD & sterile supply	Full sterilisation cycle tracking for reusable instruments
Biomedical engineering	Equipment maintenance schedules and service history
Facility & waste	Facility checks and waste handling records
Ambulance & fleet	Vehicle status and fleet management
Catering, laundry & mortuary	Support services often left out of hospital software entirely
Visitor & security	Visitor sign-in and tracking

10. Academic

For teaching hospitals: training programme rosters, clinical logbooks, CME tracking, a research project registry, and a full ethics committee review workflow.

The ethics workflow is server-enforced, not just a form. *A reviewer comment is required to record a decision — the field is explicitly marked required and the submission is rejected without it. Once a decision is finalised, it cannot be reopened; a genuine correction requires a new submission that references the original, the same amendment principle used for clinical notes.*

11. Public health

Disease surveillance with outbreak-signal detection, community outreach tracking, the full National Programme on Immunization schedule tracked per child, and national reporting formats (IDSR, NHMIS).

12. Specialty services

Seven distinct clinical and support services, each with its own workflow — grouped together because they serve a smaller, specific population rather than the whole hospital.

Service	What it's for
Nutrition & dietetics	Dietary assessment and nutrition care planning
Sickle cell centre	Ongoing management for a chronic condition requiring continuity of care
Dental & oral health	A distinct clinical workflow from general outpatient care
Infection prevention & control	Tracking and containment measures, hospital-wide
Medical social services	Social work support for patients — discharge planning, welfare concerns
Occupational health	Hospital staff's own health surveillance, separate from patient care

Service	What it's for
Chaplaincy & pastoral care	Spiritual and pastoral support as a tracked service, not an informal add-on

13. Intelligence

Analytics & KPIs, Forecasting, and Reports — the hospital's own operational picture, distinct from AgoroX's platform-side Settlement view (Section 4.3 of the Platform Admin Guide, not visible to hospital staff).

Forecasting is honestly labelled. *Forecasts are statistical projections intended to surface direction — whether something is trending up or down — not to be read as precise predicted values. Treat them as a planning signal, not a committed number.*

14. Compliance

Compliance & accreditation, Incident & risk management, and Policies & SOPs — the hospital's regulatory and governance layer.

14.1 Licenses and accreditation

Practitioner licenses and facility accreditations are both tracked with a real expiry date and status — current, expiring soon, or expired — computed live from that date, not a manually maintained flag someone has to remember to update.

14.2 Incidents and corrective action

An incident investigation requires both a corrective action and a named owner before it can be recorded — the form is explicitly rejected without both. This isn't a formality: it means every closed incident has a real, attributable action on file, not just a description of what went wrong.

15. Making HospitalOS your own

15.1 Setting your own prices

Administration → Pricing lets you override the catalogue default for any test, drug, imaging study, procedure, or accommodation tier. Every billing calculation and every price shown to staff reads through the same value.

Past charges are never rewritten. *A patient billed before you changed a price keeps their original total; only new transactions use the new price.*

15.2 Your branding

Administration → Settings has your hospital's real name, logo, address, phone, and email. All appear live across the app and on every printed document — saving here updates them everywhere within seconds, with no reload needed.

15.3 Your documents

Administration → Documents & templates accepts real uploads — drag a file in, assign any category you like, and it is immediately downloadable, renameable, and deletable. Files are stored for real: they survive a reload, a different device, or a different browser session.

16. The audit trail

Administration → Security & audit is append-only by construction: entries are frozen the moment they are written, with no update or delete capability at all. Each entry is hash-chained to the one before it, so editing a past record breaks verification at that record, and deleting one breaks verification at the next. The chain lives in the database, not the browser.

Security note. *This is tamper-evident, the honest and standard claim for this kind of log: any tampering after the fact becomes detectable. It does not claim to make database-level tampering physically impossible for someone with that level of access — no audit design does.*

17. Getting help

Press Ctrl+K (Cmd+K on Mac) from anywhere to search screens, patients, and the full help library at once. Help & documentation is pinned at the bottom of the sidebar, outside the workflow groups, reflecting that it is a standalone reference.

17.1 Downloading this guide

This guide is available at any time from Help & documentation — click “Download the full Tenant Guide” on the landing page.

17.2 The Academy

A professional certification framework — five credentials across four tiers, from HospitalOS Certified User (the entry point for every role) through Professional, Advanced, and Expert/Leadership certifications aligned to real HospitalOS workflows. It will be a paid feature and stays genuinely locked until the full curriculum is built, rather than opening early as an unfinished preview. The certification blueprint — objectives, competency areas, and proposed structure for every credential — already exists as a downloadable document from the Academy screen today, ahead of the interactive curriculum.

18. Troubleshooting common situations

Exact messages you'll actually see, and what they mean — not a generic FAQ, but the real server responses for the situations staff hit most often.

What you see	What's actually happening
“Open a cash session before taking a cash payment.”	Cash specifically requires an open session; every other method doesn't. Open one first.
“You already have an open cash session. Close it before opening a new one.”	Only one session per cashier at a time — close the existing one before starting another.
“Only approved claims can be marked paid.”	The claim lifecycle is server-enforced — approve it first, the button won't skip the step.
“This payment has ₦X left to refund; ₦Y exceeds that.”	A payment can't be refunded beyond what's genuinely left on it, even across multiple partial refunds.
“Only N unit(s) in stock.”	The server checked real, current stock at the moment of the request — not a possibly-stale number the screen was showing.

What you see	What's actually happening
“Not signed in, or your session has expired.”	Your session ended — either it timed out, or you (or an administrator) signed it out. Sign in again to continue.
A button or whole sidebar group is simply missing	Almost always a role boundary, not a fault — check the role's area/grant list under Administration → Users & roles before assuming a bug.

19. Current state, stated honestly

An honest account of what genuinely isn't built yet. Everything not listed here — including every module and workflow described above — is real and persists.

- The instruments gateway simulates live device listening (MLLP/DICOM/print daemon); device management and monitoring are real.
- No real SMS/WhatsApp/email carrier is connected yet; the message queue, templates, and channel selection are real, but delivery stays at “queued.”
- Revenue mix by module in Settlement is only real for payments taken against a generated invoice; a direct payment can't be traced back to a specific module and shows as unattributed.
- There is no self-service “forgot password” flow — your hospital's own administrator resets access.

Online payments are live. *Card / online checkout runs through a real payment provider (Paystack) on this deployment, with every online payment signature-verified and independently re-confirmed with the provider before it posts — see Section 8.2. This replaces the earlier state in which no gateway was connected.*

Questions about this guide? *support@agorox.africa*